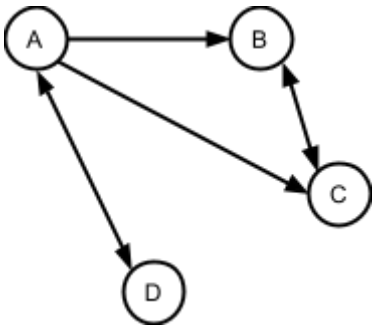
American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Senior Division

1. Graph Theory

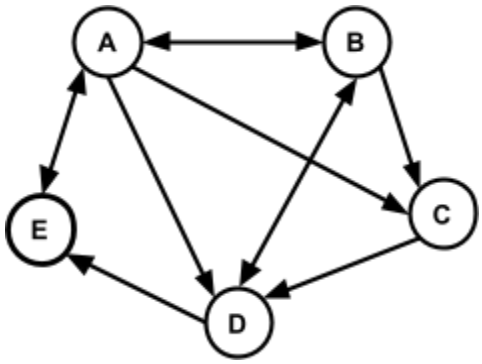
What is the shortest path length that will have more than one path between two vertices in the following directed graph?



A. 1
B. 2
C. 3
D. 4
E. 5

2. Graph Theory

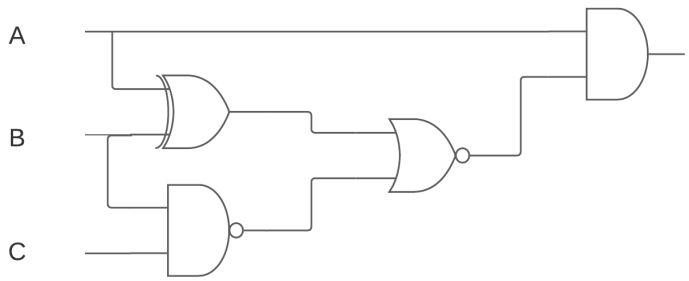
How many cycles are there in the following directed graph?



A. 5
B. 7
C. 9
D. 10
E. 12

3. Digital Electronics

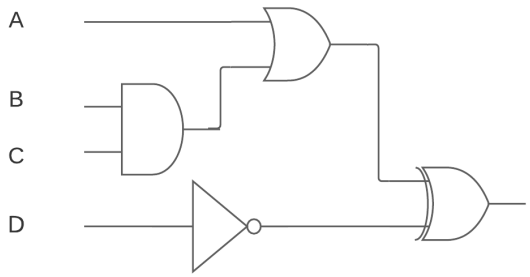
Simplify the Boolean expression of this digital circuit.



A. $A\,B\,C$
B. $\overline{A}\,B\,C$
C. $A\,\overline{B}\,C$
D. $A\,B\,\overline{C}$
E. $\overline{ABC}$

4. Digital Electronics

How many ordered quadruples make the following digital circuit TRUE?



A. 0
B. 4
C. 8
D. 12
E. 16

5. Assembly Language

What is printed when the following program is executed?

```
N     DC      5
      LOAD    N
      STORE   F
      SUB     =1
TOP   STORE   B
      LOAD    F
      MULT    B
      STORE   F
      LOAD    B
      SUB     =1
      BE      OUT
      LOAD    B
      SUB     =1
      BU      TOP
OUT   PRINT   F
      END
```

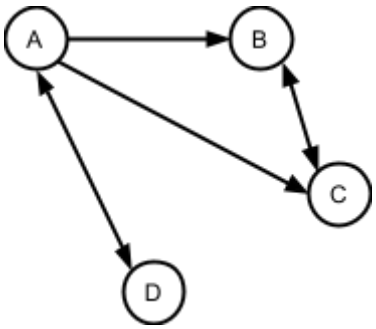
A. 15
B. 30
C. 60
D. 120
E. 240

American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Senior Division

1. 图论

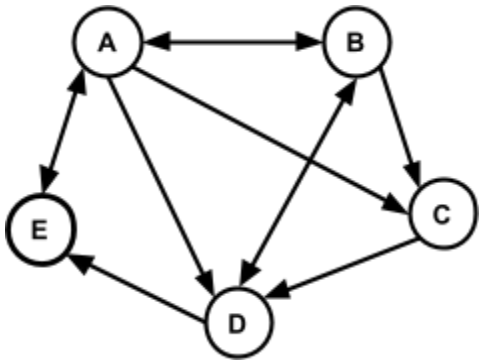
在下述有向图中，求任意两个顶点之间存在的不唯一路径的最短路径长度是多少？



A. 1
B. 2
C. 3
D. 4
E. 5

2. 图论

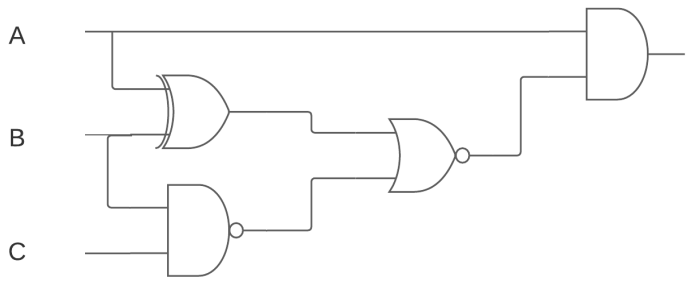
在下述有向图中有多少个环？



A. 5
B. 7
C. 9
D. 10
E. 12

3. 数字电路

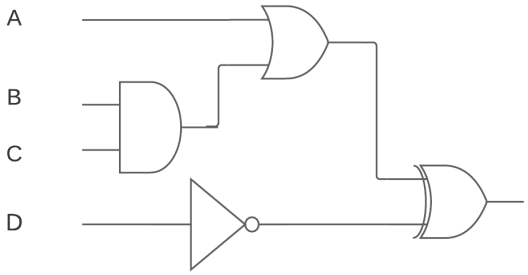
化简下述数字电路表示的布尔表达式。



A. $A\,B\,C$
B. $\overline{A}\,B\,C$
C. $A\,\overline{B}\,C$
D. $A\,B\,\overline{C}$
E. $\overline{ABC}$

4. 数字电路

有多少个有序四元组能够使得下述数字电路为**真**？



A. 0
B. 4
C. 8
D. 12
E. 16

5. 汇编语言

执行以下程序会打印输出什么？

```
N     DC      5
      LOAD    N
      STORE   F
      SUB     =1
TOP   STORE   B
      LOAD    F
      MULT    B
      STORE   F
      LOAD    B
      SUB     =1
      BE      OUT
      LOAD    B
      SUB     =1
      BU      TOP
OUT   PRINT   F
      END
```

A. 15
B. 30
C. 60
D. 120
E. 240